

NVMe® Technology: The Command Set for High Bandwidth AI, Cloud and Enterprise Applications

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Agenda

- Update on State of NVM Express
- Discuss NVM Express Activities
- Announce the Latest NVM Express® (NVMe®) Specification Set Versions
- Highlight New Features and Capabilities
- NVM Express Feature Roadmap

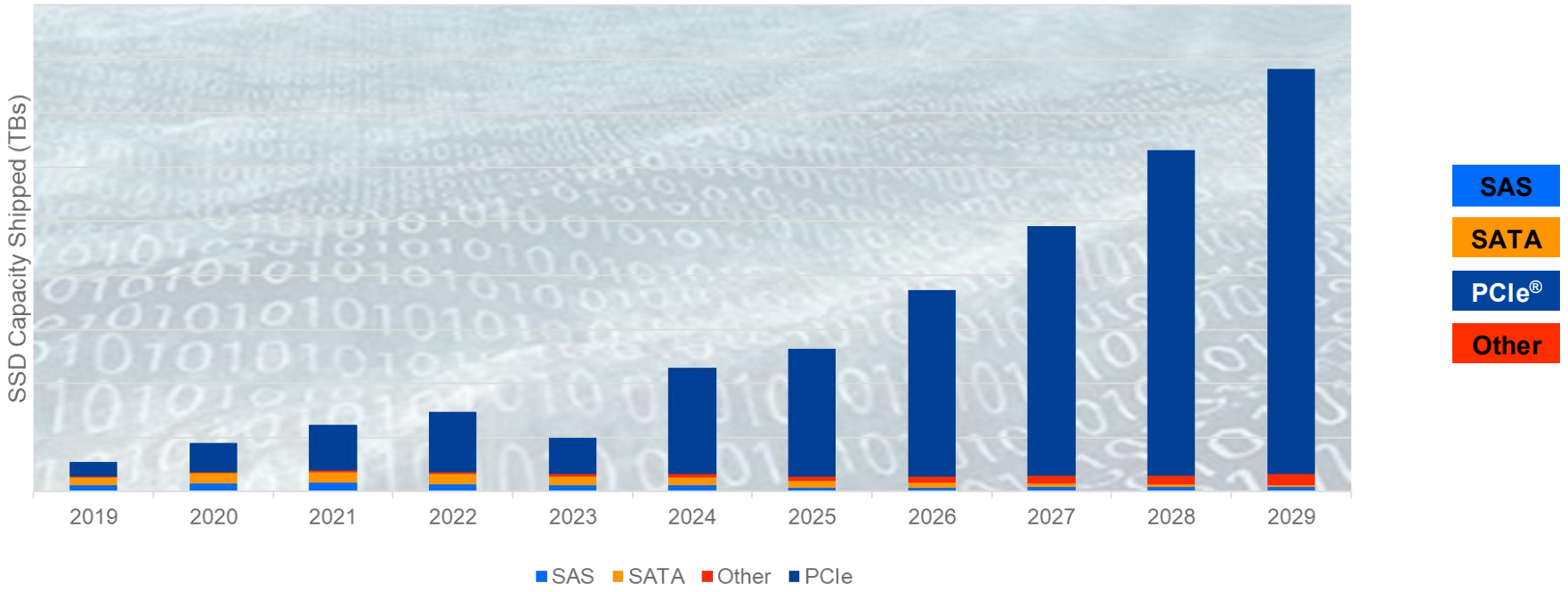
NVM Express, Inc. Overview

- NVM Express is **110+ members** strong and was created to expose the benefits of non-volatile memory in all types of computing environments. NVM Express develops the open collection of the NVMe set of specifications
- NVMe technology delivers high bandwidth, low latency storage and overcomes bottlenecks
- Markets enhanced by NVM Express technology include:
  - Artificial Intelligence
  - Virtual Reality
  - Machine Learning
  - Cloud/Data Center
  - SSD Controllers
  - Storage
  - PC/Mobile/IoT
  - Healthcare

Promoter Group
2025 - 2026

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DELL Technologies
Google
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micron
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Enterprise SSD Capacity Shipment by Interface



Source: *IDC Worldwide Solid State Drive Forecast Update, 2025–2029,Doc # US53972625 , Dec 2025*

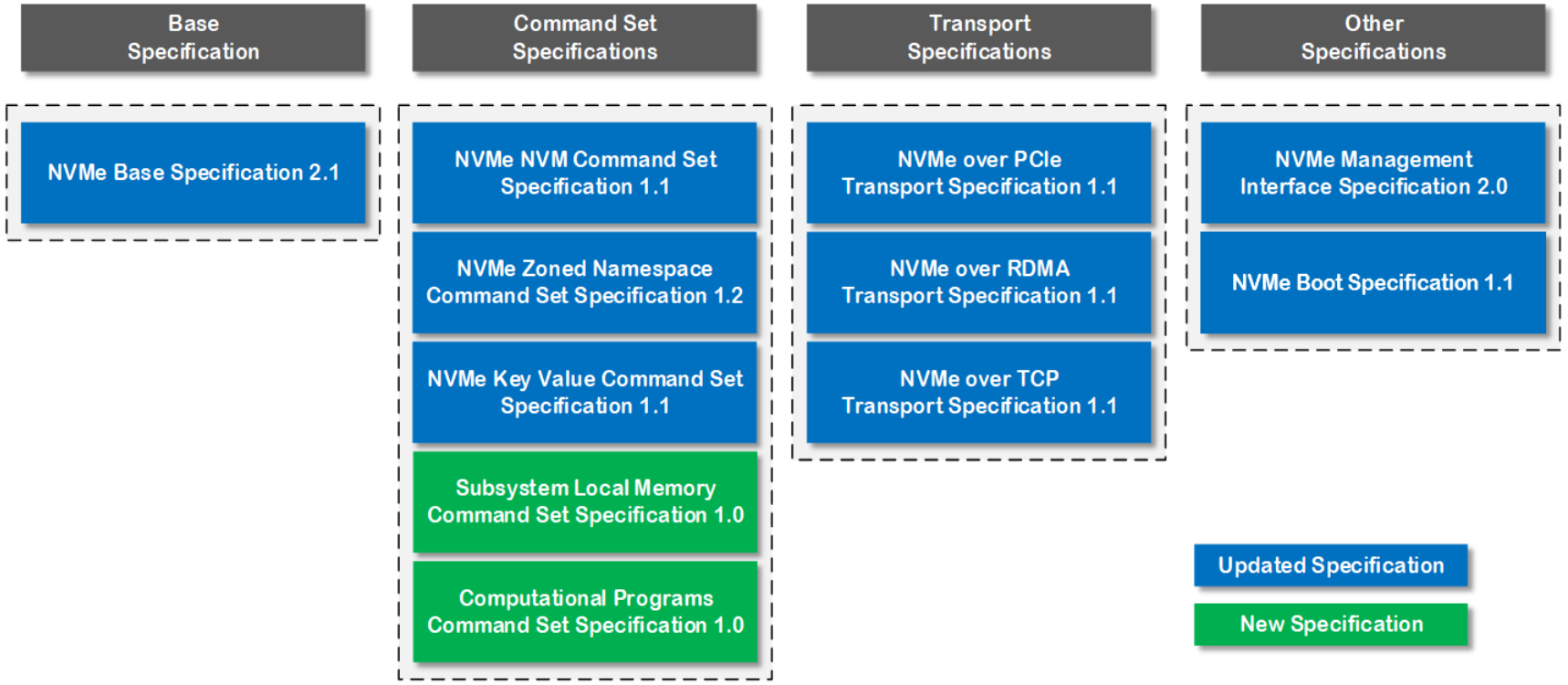
New NVMe® 2.4 Set of Specifications

On Aug. 4, 2026, NVM Express published their yearly NVMe 2.4 Set of Specifications – meeting the goal of streamlining development and time to market.

New Work Items	Ratified Technical Proposals	Ratified ECNs
17	15	11

NVM Express® architecture and its features have evolved to serve as the common architecture for client, cloud, AI, and enterprise storage.

NVMe® 2.4 Set of Specifications (August 2026)



All 11 Specifications Updated

New NVMe® 2.4 Features and Enhancements



Security

- NVMe Information Package for TDISP (TP4213)
- Lockdown Persistence (TP4200), Lockdown for Virtualization (TP4222)
- Enhancements for PQC (TP4201)



Sustainability

- Voltage Monitoring (TP4210)
- Restore Manufacturing Default Settings (TP4195)
- Idle I/O Exist Latency Limit (TP4204)



Management

- PCIe Exported NVM Subsystem Migration (TP4193)
- Rate Limiting (QoS) (TP4176)
- Host Addressable SLM (TP4184), Add RU Scope Reporting (TP4215)
- NVMe-MI Status Reporting (TP6039), NVMe-MI Persistent Data Area (TP6040)
- SFF E2 Form Factor (TP6041), CIAP & Error clarifications (TP6042)



Clarifications

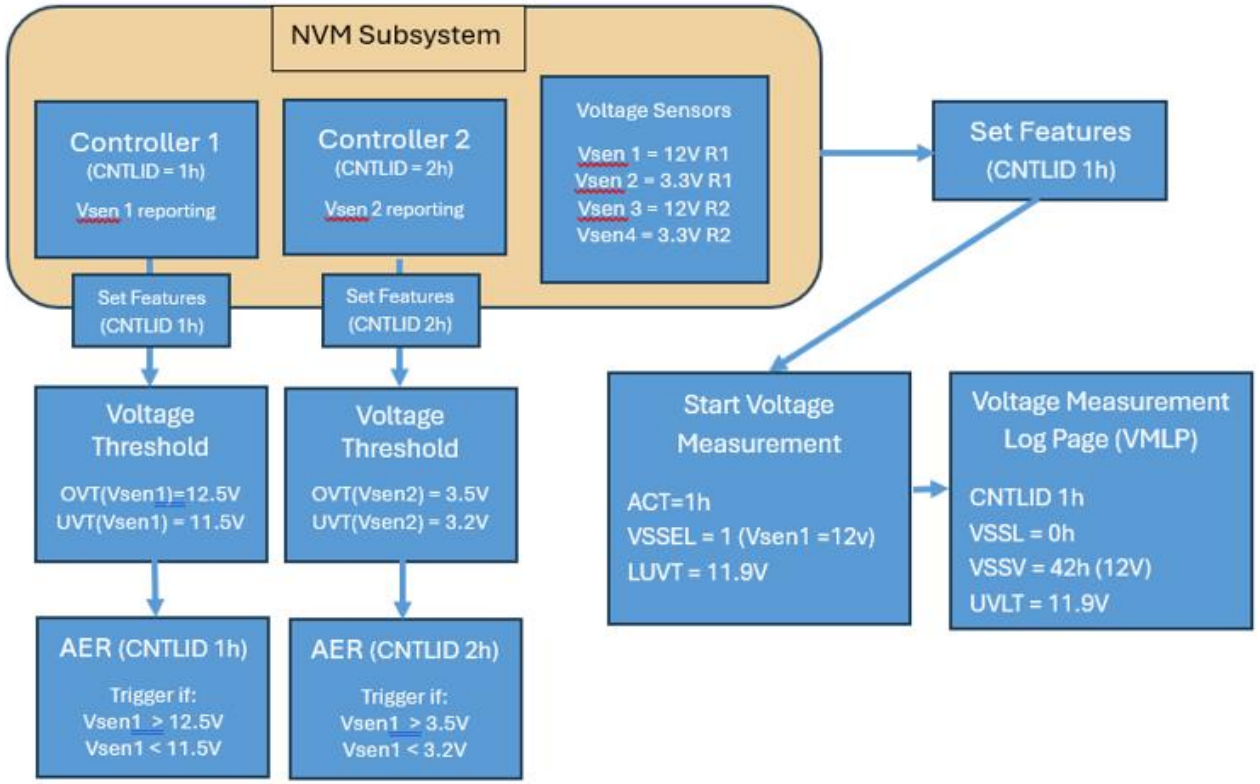
- Obsolete Predictable Latency Mode and NVM Sets (TPAR9000 & TPAR9001)
- ECN131, ECN132, ECN133, ECN134, ECN135, ECN136,
- ECN137, ECN138, ECN139, ECN140, ECN141

Security Related Updates

- NVM Express® over Fabrics (NVMe-oF™) Authentication
  - Add PQC algorithms
- PCIe® TDISP Enablement (Attestation)
  - Prove it's the "same" controller
- Command Lockdown for Virtualization
  - Expand lockdown for virtual controllers
- Lockdown Persistence Across Power Cycle
  - Allow command lockdown in more use cases

Sustainability Updates

- Voltage Monitoring (TP4210)
  - Track/Monitor up to 4 voltage rails within the subsystem



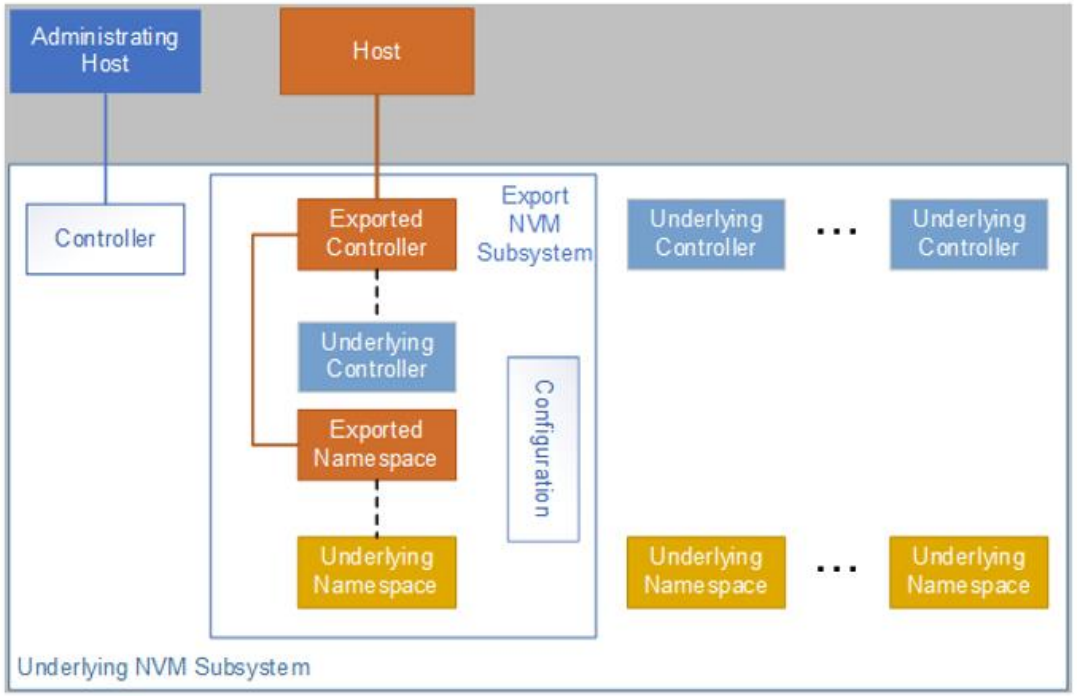
Sustainability Updates (cont.)

- Restore Manufacturing Default Settings
  - Get the device back to its original state
  - Promote the circular economy (reuse)

- Idle I/O Exist Latency Limit
  - Minimum power without a long power up delay

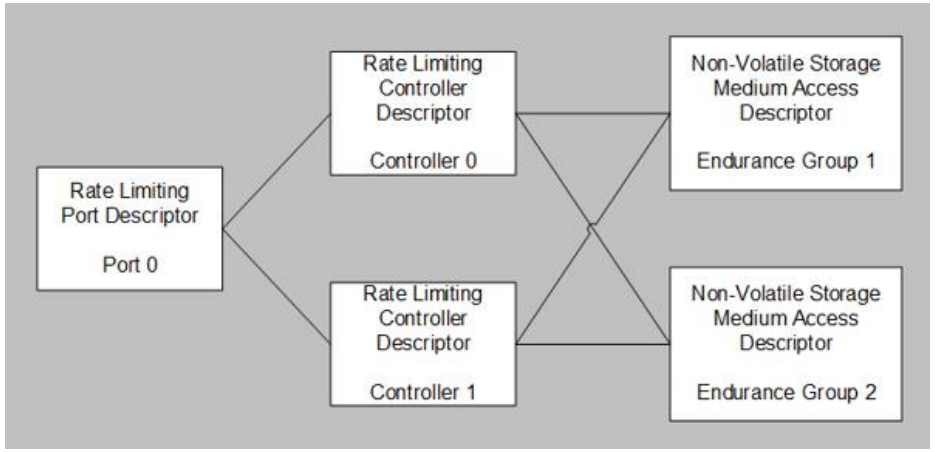
New Features

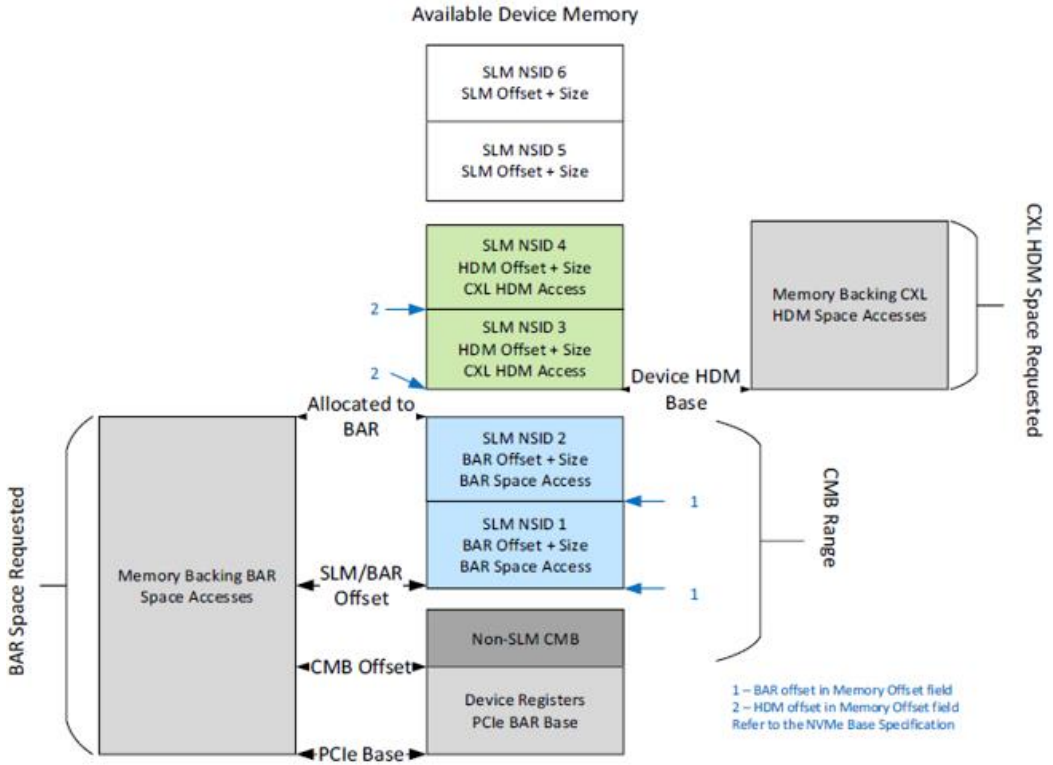
- PCIe® Exported NVM Subsystem Migration
  - Enable exported subsystem (virtual subsystem) operation on PCIe technology
  - Enable migration of virtual subsystems



New Features (cont.)

- Host Addressable SLM
  - Enable SLM NS to have memory addresses
- Rate Limiting (QoS)
  - Fairness between migration load and “real” work

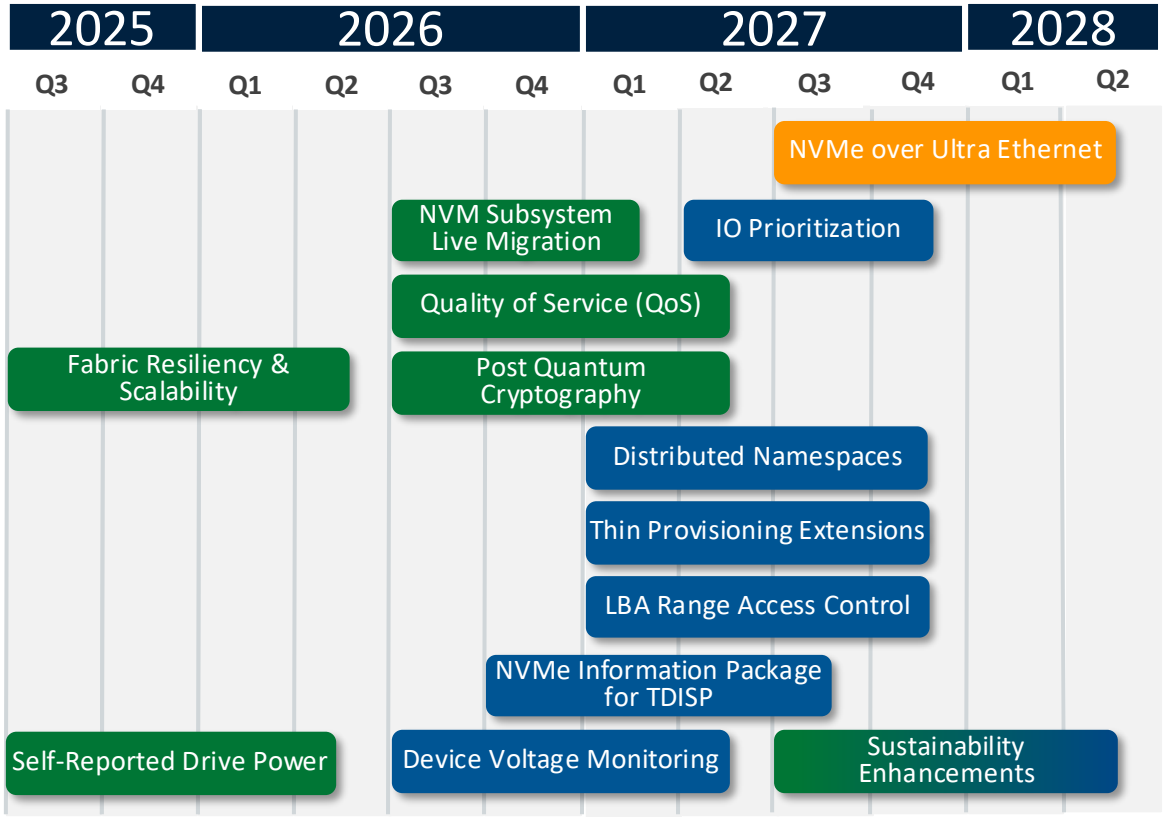




NVMe® Management Interface (NVMe-MI™) Architecture Updates

- Status Reporting Enhancements
- Persistent Data Area for MI
- Add SFF E2 Form Factor
- CIAP Enhancements & Header Error Clarifications

NVMe® Planned Feature Roadmap



Ratified Feature
(left edge indicates ratification quarter)

Planned Feature
(left edge indicates planned ratification quarter)

New Specification (left edge indicates planned ratification quarter)

Updated August 2026

Summary

- NVMe® technology adoption continues to grow and has succeeded in unifying client, cloud, AI, and enterprise storage around a common architecture
  - Systems are being built using NVMe technology as the native underlying storage technology for cell phones, tablets, laptops, desktops, storage arrays, data centers, automotives, etc.
- A dedicated NVMe technical community maintains existing specifications while developing new innovations; since 2.3 release, we have had:
  - 17 new work items authorized
  - 15 Technical Proposals ratified
  - 11 ratified ECNs
- NVMe technology began as short and simple PCIe SSD specification and has grown into nearly a dozen specifications supporting all major transports, multiple command sets, and standardizing many aspects of storage
  - Technical Proposals are made public when ratified
  - Specification updates simply aggregate Technical Proposals and ECNs ratified since the last specification update

Call to Action

- Join NVM Express:
  - We invite you to join NVM Express and contribute to the development of future NVM Express® (NVMe®) technology features needed by the OCP community
  - Visit nvmexpress.org/join-nvme/ to learn more
- For More Information:
  - View the latest NVMe set of specifications: nvmexpress.org/specifications/
  - View the NVM Express educational resources
    - NVM Express blog: nvmexpress.org/category/blog/
    - NVM Express educational resources: nvmexpress.org/education/
  - Follow NVM Express on social media for the latest updates

Thank you!

